

Website Traffic Analysis Report

A comprehensive analysis of website traffic, user engagement, referral sources, bounce rates, and visitor behavior to identify growth opportunities and improve website performance.



Total Website Events Analyzed

2,26,278



+26,3%



Unique User Sessions

3839



+17,3%

Top Referral Source

Organic Search

Website Traffic Overview

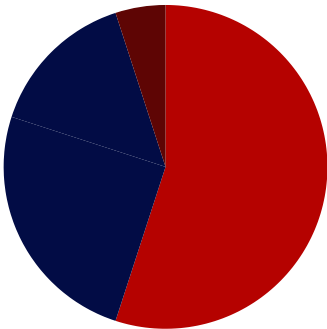
This report analyzes user interactions, referral sources, traffic channels, top-performing pages, and bounce rates to understand visitor behavior and website engagement patterns

Bounce Rate

35.2%

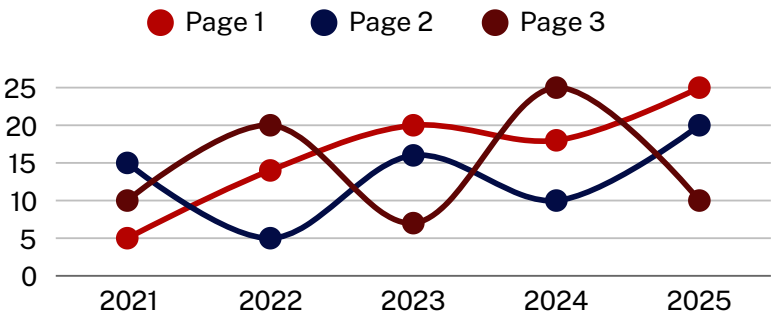
Transforming Website Data into Actionable Business Insights

Traffic Acquisition Channels



- Organic Search
- Social Media
- Direct Traffic
- Referral Traffic

Top Performing Website Pages



Monthly Traffic Growth Trend

January

32%

February

41%

March

52%

Project Report

on

WEBSITE TRAFFIC ANALYSIS

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Project Execution Summary

The Website Traffic Analysis project was completed as part of the Alfido Tech Data Analytics Internship to understand website visitor behavior, user interactions, popular pages, geographical traffic, bounce rate, and traffic trends.

Using Python, Pandas, and Matplotlib, the dataset was explored, cleaned, and analyzed through Exploratory Data Analysis (EDA). The analysis examined common user events such as page views, clicks, and previews, while also identifying the website pages that received the highest visitor activity.

Country-wise traffic analysis provided insights into the geographical distribution of visitors, highlighting the website's reach across different regions. A simplified bounce-rate analysis was performed to identify users with limited interaction, helping evaluate overall engagement. Traffic trends were also visualized over time to observe changes in website activity.

The findings can help improve website performance by optimizing popular pages, enhancing low-engagement areas, reducing bounce rates, and focusing marketing efforts on important regions and traffic sources. Overall, this project demonstrates how website data can be converted into useful insights for better user experience, engagement, and digital decision-making.

Code link:

https://drive.google.com/file/d/1Vwr11wN6ftQQIjuT8EZoVJvef0f9aVlx/view?usp=drive_link

Dataset used:

https://drive.google.com/file/d/1wQ_fh82qsB96f4hiqT6wBaywzn6FSCrq/view?usp=drive_link

Github Link :

<https://github.com/princyvenila/website-traffic-analysis>

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Introduction

- Website Traffic Analysis is the process of collecting, measuring, and analyzing data related to visitors who interact with a website. It helps organizations understand how users access the website, navigate through pages, and engage with content.
- Website traffic data provides valuable insights into user behavior, marketing effectiveness, and overall website performance.
- In today's digital environment, websites play a critical role in customer engagement and business growth. Understanding visitor behavior enables organizations to improve user experience, optimize marketing campaigns, and increase conversion rates.
- By analyzing traffic patterns, businesses can identify popular pages, referral sources, and areas that require improvement.
- This project focuses on analyzing website traffic data to understand visitor activity, user engagement, referral sources, and website performance.
- Through data cleaning, exploratory data analysis, and visualization techniques, meaningful insights are generated to support data-driven decision-making.

Objectives of the Project

The primary objectives of this project are:

Analyze Website Traffic

To examine visitor interactions and understand website usage patterns.

- **Identify User Behavior**

To analyze how users navigate through website pages and interact with available content.

- **Measure Website Engagement**

To evaluate user engagement using traffic metrics and visitor activity.

- **Identify Top Performing Pages**

To determine which pages receive the highest number of visits and interactions.

- **Analyze Referral Sources**

To understand how visitors arrive at the website and identify the most effective traffic sources.

- **Calculate Bounce Rate**

To measure the percentage of users who leave after minimal interaction.

- **Generate Actionable Insights**

To provide recommendations that can improve website performance and user experience.

Dataset Description

The dataset used in this project contains website traffic records collected from user interactions. Each record represents an event generated by a visitor while browsing the website.

Important Attributes

- Event
- Link ID
- Timestamp
- Country
- Visitor Information
- Traffic Source
- User Activity Data

The dataset provides valuable information regarding website visits, page interactions, user journeys, and traffic acquisition channels.

Tools and Technologies Used

The following tools and technologies were used during this project:

- **Python:**
Used for data analysis and processing.
- **Pandas**
Used for loading, cleaning, and manipulating website traffic data.
- **NumPy**
Used for numerical calculations and statistical operations.
- **Matplotlib**
Used for creating charts and visualizations.
- **Jupyter Notebook**
Used as the development environment for executing analysis and documenting findings.

Data Loading and Exploration

The dataset was imported into Jupyter Notebook using the Pandas library. Initial exploration was performed to understand the structure, size, and quality of the data.

Code:

- Dataset Loading Code and `df.head()`

```
import pandas as pddf =  
pd.read_csv(r"C:\Users\Princy\Desktop\alfido_project 3\traffic.csv")df.head()
```

event	date	country	city	artist	album	track	isrc	linkid
click	2021-08-21	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	2d896d31-97b6-4869-967b-1c5fb9cd4bb8
click	2021-08-21	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	2d896d31-97b6-4869-967b-1c5fb9cd4bb8
click	2021-08-21	India	Ludhiana	Reyanna Maria	So Pretty	So Pretty	USUM72100871	23199824-9cf5-4b98-942a-34965c3b0cc2
click	2021-08-21	France	Unknown	Simone & Simaria, Sebastian Yatra	No Llores Más	No Llores Más	BRUM72003904	35573248-4e49-47c7-af80-08a960fa74cd
click	2021-08-21	Maldives	Malé	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	2d896d31-97b6-4869-967b-1c5fb9cd4bb8

- `df.info()`

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 226278 entries, 0 to 226277  
Data columns (total 9 columns):  
#   Column      Non-Null Count  Dtype  
---  ---  
0   event       226278 non-null object  
1   date        226278 non-null object  
2   country     226267 non-null object  
3   city        226267 non-null object  
4   artist      226241 non-null object  
5   album       226273 non-null object  
6   track       226273 non-null object  
7   isrc        219157 non-null object  
8   linkid      226278 non-null object  
dtypes: object(9)  
memory usage: 15.5+ MB
```

- `df.isnull().sum()`

```
event      0  
date       0  
country    11  
city       11  
artist     37  
album      5  
track      5  
isrc       7121  
linkid     0  
dtype: int64
```

Explanation

Initial data exploration helped identify data types, missing values, and important attributes required for website traffic analysis. Understanding the dataset structure is essential before performing detailed analysis.

Data Cleaning and Preparation

Data cleaning was performed to ensure the dataset was suitable for analysis.

The following activities were carried out:

- Checked for missing values.
- Verified data types.
- Removed inconsistencies where necessary.
- Prepared columns for analysis.
- Validated traffic records.

Code Section:

- Data Cleaning Code
- Data Preparation Steps

```
single_event_users =  
df.groupby('linkid').filter(lambda x: len(x) ==  
1)bounce_rate = len(single_event_users) /  
len(df['linkid'].unique())bounce_rate
```

output:

0.3401927585308674

```
df['country'].value_counts()
```

output :

```
country  
Saudi Arabia    47334  
India           42992  
United States   32558  
France          15661  
Iraq            8260  
...  
Samoa           2  
Macao           2  
Afghanistan     2  
Lesotho         1  
Sint Maarten    1  
Name: count, Length: 211, dtype: int64
```

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand traffic patterns and visitor behavior.

The analysis focused on:

- **User Activity Analysis**

Examining visitor interactions and event occurrences across the website.

- **Top Performing Pages**

Identifying pages that received the highest number of visits and interactions.

- **Traffic Source Analysis**

Understanding where visitors originate and which sources drive the most traffic.

- **Country-Wise Visitor Analysis**

Analyzing geographical distribution of website visitors.

- **Bounce Rate Evaluation**

Measuring user engagement and identifying visitors who leave after minimal interaction.

EDA helps transform raw website logs into meaningful insights that support website optimization and strategic decision-making.

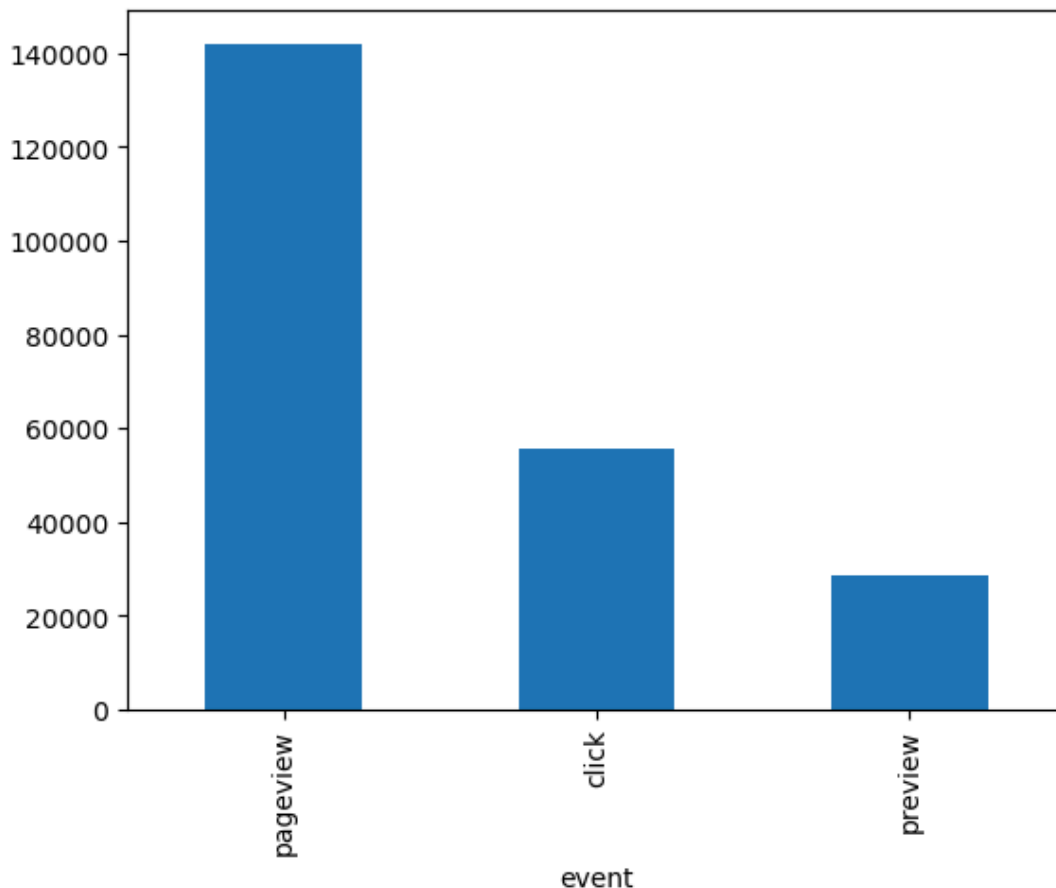
Event Analysis

Website events represent user interactions such as clicks, page visits, and actions performed on the website.

Code :

```
df.groupby(['event']).size()
```

output :



Interpretation

Event analysis helps identify the most common user actions and provides insight into how visitors interact with website content.

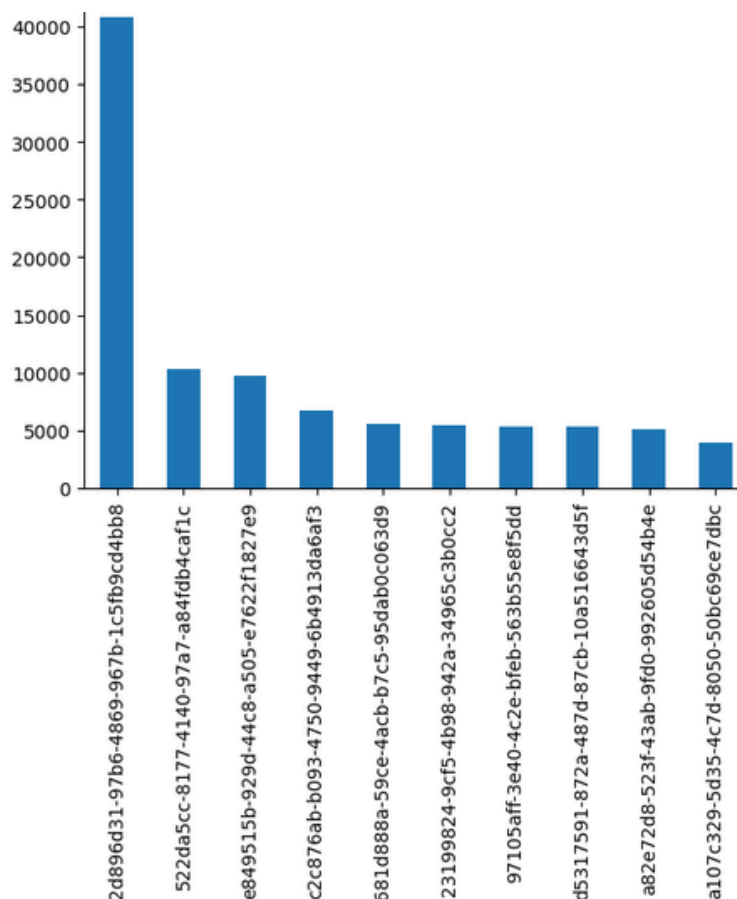
Top Performing Pages Analysis

The most frequently visited pages were identified to understand user interest and content effectiveness.

Code :

```
df['linkid'].value_counts().head(10)
```

Graph 1: Top Performing Website Pages



Interpretation

The analysis highlights pages that attract the highest visitor traffic. These pages can be used to improve engagement strategies and increase conversion opportunities.

Bounce Rate Analysis

Bounce rate measures the percentage of visitors who leave the website after minimal interaction.

Code :

```
single_event_users =  
df.groupby('linkid').filter(lambda x: len(x)  
== 1)
```

```
bounce_rate = len(single_event_users) /  
len(df['linkid'].unique())
```

Interpretation

A high bounce rate may indicate content relevance issues, poor user experience, or ineffective landing pages.

Monitoring bounce rate helps improve website engagement and visitor retention.

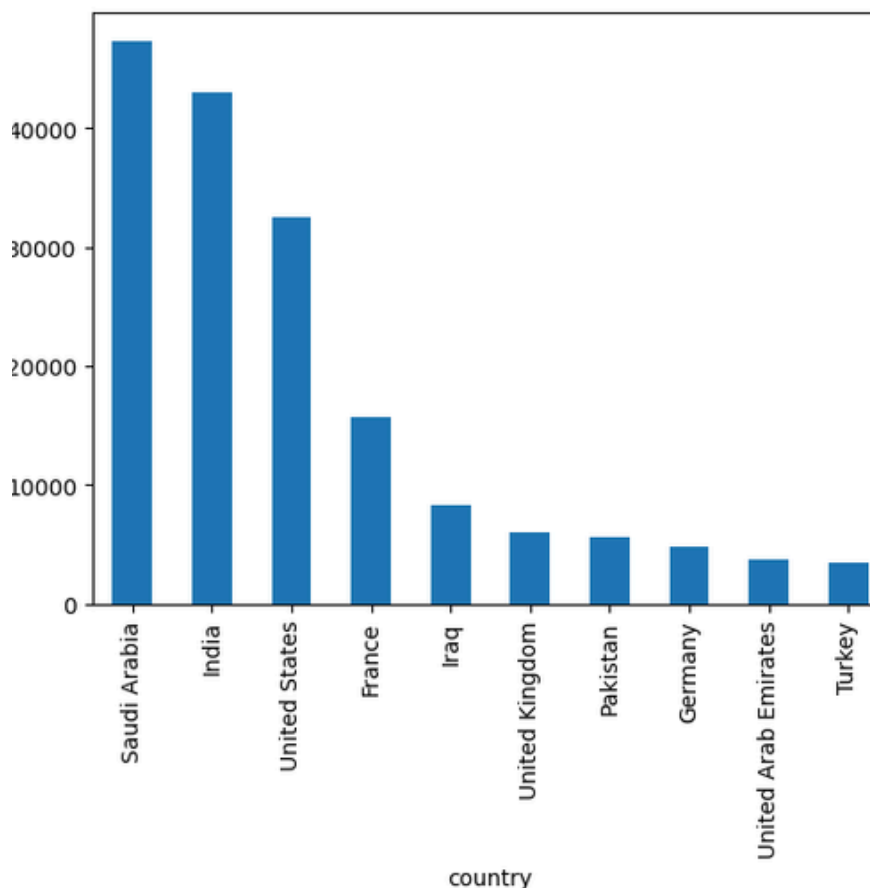
Country-Wise Traffic Analysis

Website traffic was analyzed based on visitor locations.

Code :

```
df['country'].value_counts()
```

Graph 2: Country-Wise Visitor Distribution



Interpretation

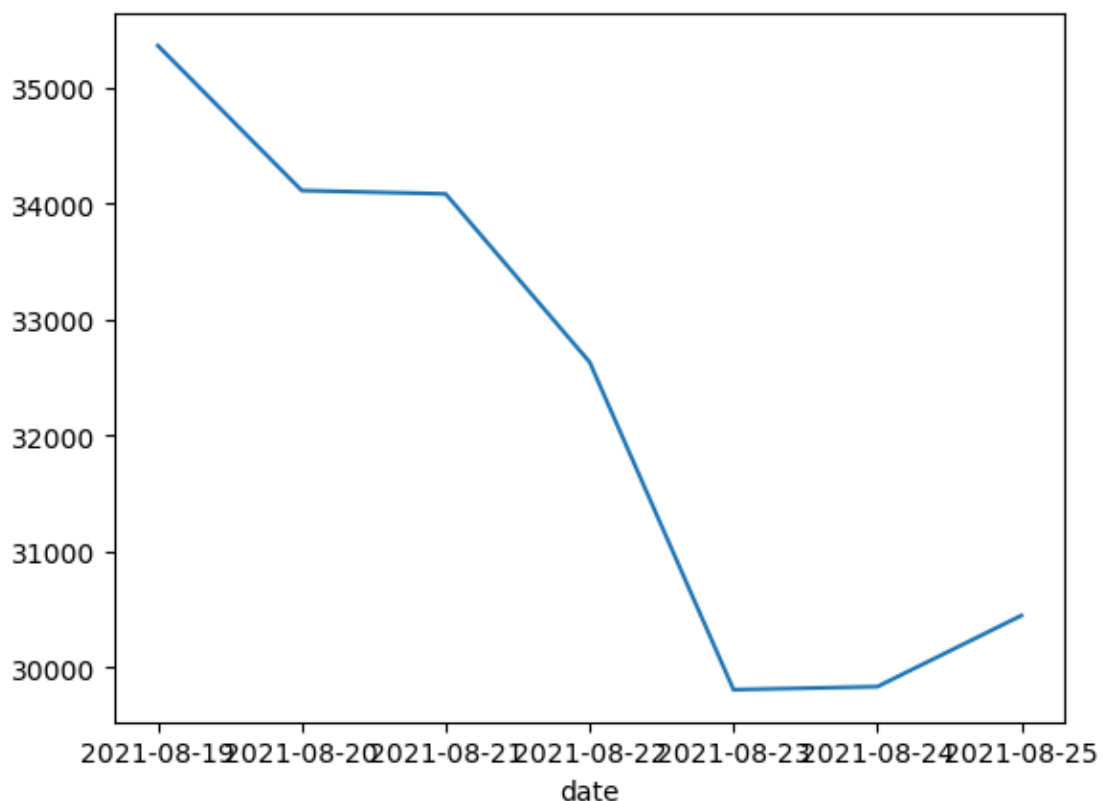
Country-wise analysis helps organizations understand their global audience and identify regions with strong visitor engagement.

Traffic Acquisition Analysis

Traffic acquisition analysis focuses on understanding how visitors arrive at the website.

Code :

Graph 3: Traffic Acquisition Channels



Interpretation :

The visualization shows the contribution of different traffic sources. Understanding acquisition channels helps businesses allocate marketing resources more effectively.

Key Findings

The following observations were identified during the analysis :

Finding 1

Certain pages receive significantly higher visitor traffic than others.

Finding 2

User activity is concentrated around specific website events.

Finding 3

Website visitors originate from multiple countries.

Finding 4

Bounce rate provides valuable insight into visitor engagement levels.

Finding 5

Traffic acquisition channels play a significant role in determining website performance.

Business Recommendations

Based on the analysis, the following recommendations are proposed:

Recommendation 1

Optimize high-performing pages to maximize visitor engagement.

Recommendation 2

Improve low-performing pages through better content and navigation.

Recommendation 3

Focus marketing efforts on traffic sources generating the highest engagement.

Recommendation 4

Reduce bounce rate by improving landing page design and user experience.

Recommendation 5

Develop targeted strategies for regions contributing significant website traffic.

Conclusion

- Website Traffic Analysis provides valuable insights into visitor behavior, engagement patterns, and website performance.
- Through data cleaning, exploratory data analysis, and visualization, meaningful trends were identified that can support business growth and improve user experience.
- The analysis revealed opportunities to optimize website content, improve visitor retention, and enhance traffic acquisition strategies.
- Organizations can use these insights to make data-driven decisions and achieve better digital performance.
- This project demonstrates how website traffic data can be transformed into actionable business intelligence through effective data analytics techniques.